

Stage 2: Data Preparation and GARCH Volatility

Does GARCH Information Improve Forecasts of Implied Volatility?

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1 Data and sample

The study uses daily observations on the CBOE Volatility Index (VIX) and the S&P 500 Composite Index over **3 January 2012 to 21 August 2026**, giving **3,680 matched trading days**.

The VIX is taken from CBOE’s official history file and the S&P 500 from Yahoo Finance (`^GSPC`). No single free provider covers both series, since CBOE does not publish index history—the same constraint Liu et al. (2015) document. The Yahoo VIX was also downloaded and validated against CBOE’s: the two agree exactly, to two decimal places, on **99.76%** of the 3,681 common dates, with a correlation of 0.99997 and nine days differing by more than 0.01. This reproduces the conclusion of Liu et al. (2015) on more recent data; the CBOE series is nonetheless used throughout, since it is authoritative and freely available.

Calendars were compared before merging rather than letting an inner join discard mismatches silently. CBOE reports a VIX close on 32 dates on which the NYSE cash market was closed, VIX being calculated from SPX option quotes disseminated in sessions that need not coincide with equity trading hours; no matched return can be formed on those days and they are excluded. Neither series contains internal missing values or duplicate dates, and the largest gap between consecutive observations is the five-day NYSE closure for Hurricane Sandy ending 31 October 2012, left unimputed. Returns are computed as $r_t = 100 [\ln(P_t) - \ln(P_{t-1})]$ on the *full* price history before the sample is truncated, so the first in-sample return is measured against the 30 December 2011 close rather than being undefined.

2 Descriptive statistics

Table 1: Descriptive statistics, 3 January 2012 – 21 August 2026 ($N = 3,680$). Kurtosis is unstandardised (normal = 3).

	VIX	S&P 500 return (%)
Mean	17.746	0.049
Median	16.180	0.068
Standard deviation	6.471	1.051
Minimum	9.140	−12.765
Maximum	82.690	9.089
Skewness	+2.863	−0.638
Kurtosis	18.939	19.314
Jarque–Bera (p)	$< 10^{-300}$	$< 10^{-300}$

The mean of VIX exceeds its median, skewness is +2.86, and the maximum of 82.69 (16 March 2020) stands far above the minimum of 9.14: the index sits at a modest level most of the time and spikes sharply on occasion. Returns are the mirror image, with skewness of −0.64, since the large moves that drive volatility upward are predominantly falls. Kurtosis near 19 for both, against a Gaussian value of 3, indicates pronounced fat tails, and Jarque–Bera rejects normality overwhelmingly.

3 Time-series properties

Persistence of VIX. The sample autocorrelation function (Figure 4) declines slowly:

Lag	1	5	10	22	66	125	250
ACF	0.961	0.854	0.751	0.563	0.337	0.262	0.046

The autocorrelation remains outside the ± 0.032 white-noise band as far as lag 250—a full trading year—and decays hyperbolically rather than geometrically. This is the long-memory behaviour documented by Fernandes et al. (2014), whose explanation applies here: VIX is a 30-calendar-day forward-looking measure sampled daily, so consecutive observations refer to heavily overlapping windows, which mechanically induces persistence. It is also why their plain HAR specification is so hard to beat at short horizons, and why Wen et al. (2024) find R^2 still improving at a 20-day lookback.

Stationarity. The Augmented Dickey–Fuller test rejects the unit-root null decisively for VIX (ADF = -5.560 , $p = 1.5 \times 10^{-6}$) and for returns (ADF = -13.420 , $p = 4.2 \times 10^{-25}$). VIX is therefore stationary but strongly persistent: shocks do die out, but slowly.

Volatility clustering. The contrast between the autocorrelation of returns and of squared returns is decisive (Figure 5):

Lag	1	2	5	10	22
ACF of r_t	−0.116	+0.072	+0.034	−0.034	−0.078
ACF of r_t^2	+0.449	+0.469	+0.290	+0.218	+0.065

Raw returns are close to unforecastable, with small autocorrelations that alternate in sign; squared returns are not, remaining positive throughout. The direction of tomorrow’s move is essentially unpredictable, its magnitude highly predictable from recent magnitudes. Ljung–Box gives $Q(10) = 3,955$ on squared returns against 236 on raw returns, and Engle’s ARCH-LM test returns LM = 1,109 at five lags with $p \approx 10^{-237}$. These results establish the precondition for a conditional-variance model.

4 GARCH(1,1) estimation

The model estimated on S&P 500 log returns is

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

Table 2: GARCH(1,1) with Gaussian innovations, estimated by maximum likelihood with robust standard errors.

Parameter	Estimate	Std. error	<i>t</i> -statistic
ω	0.0431	0.0093	4.63
α	0.1666	0.0230	7.24
β	0.7907	0.0256	30.92
$\alpha + \beta$	0.9573		
Half-life of a variance shock	15.9 trading days		
Long-run annualised volatility	15.95		

All three variance parameters are significant at any conventional level, with $\alpha, \beta > 0$ and $\alpha + \beta < 1$, so the non-negativity and stationarity conditions hold. $\omega = 0.0431$ is the constant floor of the variance equation, implying a long-run daily variance of $\omega/(1 - \alpha - \beta) = 1.010$ and an annualised long-run volatility of 15.95—close to the sample standard deviation of returns, as it should be. $\alpha = 0.167$ is the reaction coefficient, raising today’s variance by 16.7% of the squared surprise, while $\beta = 0.791$ is the carry-over from yesterday’s variance, the most precisely estimated parameter ($t = 30.9$) and the one that dominates the dynamics.

$\alpha + \beta = 0.9573$ is the quantity of interest. It lies below unity, so variance is mean-reverting, but is high enough that shocks decay slowly: the half-life is $\ln(0.5)/\ln(0.9573) \approx 15.9$ trading days, roughly three trading weeks for half of a shock to dissipate. This is the standard result for daily equity index returns, where $\alpha + \beta$ typically lies between 0.95 and 0.99, and it mirrors the persistence already documented for VIX itself. Given return kurtosis of 19.3 the model was re-estimated with Student- t innovations: α is essentially unchanged at 0.167, persistence rises to 0.984 with $\nu = 5.48$, and BIC prefers that specification decisively (9,019 against 9,247). The persistence conclusion therefore does not depend on the distributional assumption; the Gaussian model is retained because the brief specifies a basic GARCH(1,1).

5 VIX and GARCH volatility compared

The conditional volatility series is returned in daily percentage units and is annualised as $\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$, placing it on the same scale as VIX. Figure 6 plots the two together.

VIX exceeds annualised GARCH volatility on **86.7%** of days, by an average of **3.19 volatility points**—**about 22% of the GARCH mean** (14.56 against a VIX mean of 17.75). A discrepancy this one-sided is not estimation error but the **variance risk premium**: VIX is a risk-neutral expectation extracted from option prices and embeds compensation demanded for bearing volatility risk, whereas GARCH, estimated from realised returns alone, delivers only the physical forecast. Liu et al. (2015) document the same shortfall and read it the same way, reporting their GARCH-computed VIX 20–30% below the market VIX before September 2003 and 10–13% below afterwards. The 22% here exceeds that post-2003 figure, which is the relevant comparison for a 2012–2026 sample; they also compare a one-day-ahead GARCH *forecast* of VIX rather than annualised conditional volatility. GARCH exceeds VIX on the remaining 13.3% of days, the largest overshoot—44 points on 17 March 2020—following directly from the -12.77% return of 16 March, since σ_t^2 depends on ϵ_{t-1}^2 and a single extreme return is mechanically propagated into the next day’s variance. GARCH reached an annualised 120 while VIX, already pricing an eventual return to calm, stood at 75.9.

In levels the two series move together closely: Pearson 0.833, Spearman 0.730, and 0.793 in logarithms, ranging across four-year sub-periods from 0.663 (2012–2015) to 0.889 (2020–2022) — strongest in the turbulent window, weakest in the calm stretches. In daily *changes*, however, the correlation collapses to -0.007 . The lead-lag profile resolves the apparent conflict:

k	−2	−1	0	+1	+2
$\text{corr}(\Delta\sigma_{t+k}^{\text{GARCH}}, \Delta\text{VIX}_t)$	+0.002	+0.026	−0.007	+0.410	+0.035

The correlation is near zero contemporaneously and $+0.41$ at a one-day lag. The reason is structural: σ_t is a function of information dated $t - 1$ and cannot respond to today’s news until tomorrow, whereas VIX is priced off current quotes and responds immediately. The two measures agree about the *level* of volatility but learn about *changes* one day apart. This bears directly on the project: GARCH volatility is not a proxy for VIX but reacts on a different schedule and carries a systematic risk-premium wedge. Whether that survives as *incremental* predictive power once VIX’s own history is conditioned upon is the question for the forecasting stage.

6 Conclusions

Is VIX highly persistent? Yes. First-order autocorrelation is 0.961, still 0.563 after one month and 0.337 after a quarter, with the ACF outside the white-noise band as far as lag 250 and decaying hyperbolically. It is nonetheless stationary (ADF = -5.560 , $p = 1.5 \times 10^{-6}$).

Is there volatility clustering in S&P 500 returns? Yes, unambiguously. Squared returns are autocorrelated at 0.449 at lag 1 and remain positive beyond lag 20 while raw returns are near-unforecastable; Ljung–Box gives $Q(10) = 3,955$ against 236, and ARCH-LM returns LM = 1,109 with $p \approx 10^{-237}$.

Is GARCH volatility highly persistent? Yes. $\alpha + \beta = 0.9573$ implies a half-life of 15.9 trading days, with a long-run annualised volatility of 15.95; under Student- t errors persistence rises to 0.984.

How closely do VIX and GARCH volatility move together? Closely in levels (Pearson 0.833, ranging 0.663–0.889 across four-year sub-periods), hardly at all in daily changes (-0.007). The lead-lag correlation of $+0.410$ at one day shows GARCH responding to the same information a day later, because σ_t conditions on data through $t - 1$ while VIX is priced off current quotes.

What is the main difference between VIX and GARCH volatility? They are different objects rather than competing estimates of the same one. VIX is forward-looking and risk-neutral, extracted from option prices and therefore carrying a variance risk premium—the 22% wedge—and it reprices within the trading day. GARCH volatility is backward-looking, computed mechanically from realised returns under the physical measure, and cannot move until the day after a shock.

References

- Fernandes, M., Medeiros, M. C., & Scharth, M. (2014). Modeling and predicting the CBOE market volatility index. *Journal of Banking & Finance*, 40, 1–10.
- Liu, Q., Guo, S., & Qiao, G. (2015). VIX forecasting and variance risk premium: A new GARCH approach. *North American Journal of Economics and Finance*, 34, 314–322.
- Wen, C., Zhai, J., Wang, Y., & Cao, Y. (2024). Implied volatility is (almost) past-dependent: Linear vs non-linear models. *International Review of Financial Analysis*, 95, 103406.

Appendix: Figures

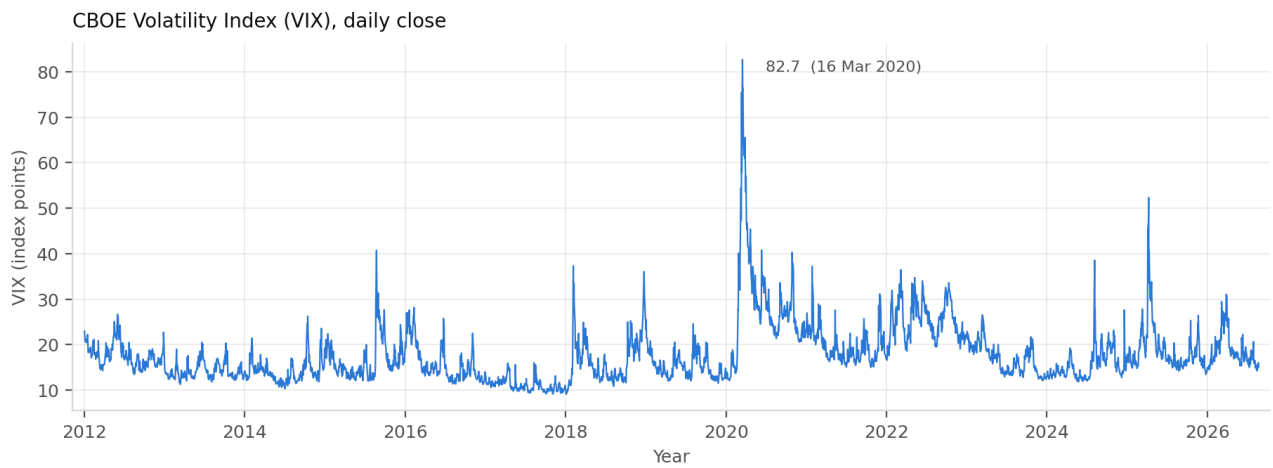


Figure 1: VIX, daily close, 2012–2026. Source: CBOE.

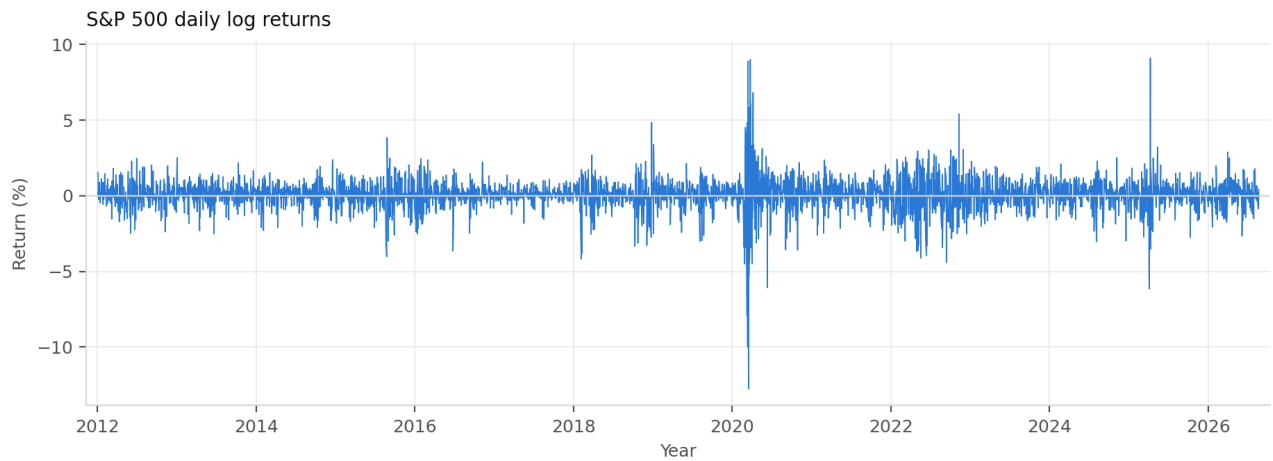


Figure 2: S&P 500 daily log returns. Amplitude varies systematically over time.

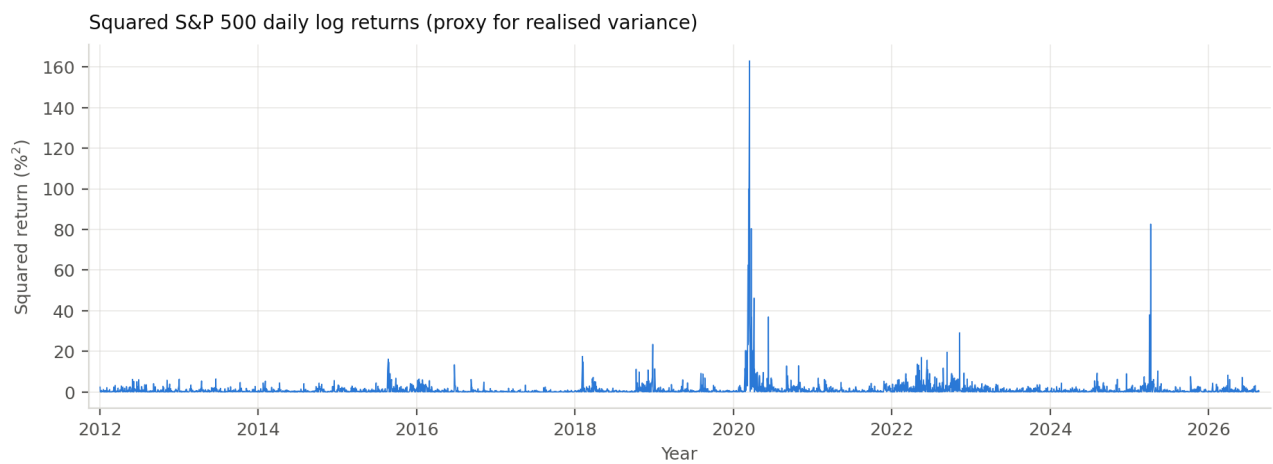


Figure 3: Squared S&P 500 daily log returns. Removing the sign makes the clustering of large moves unmistakable.

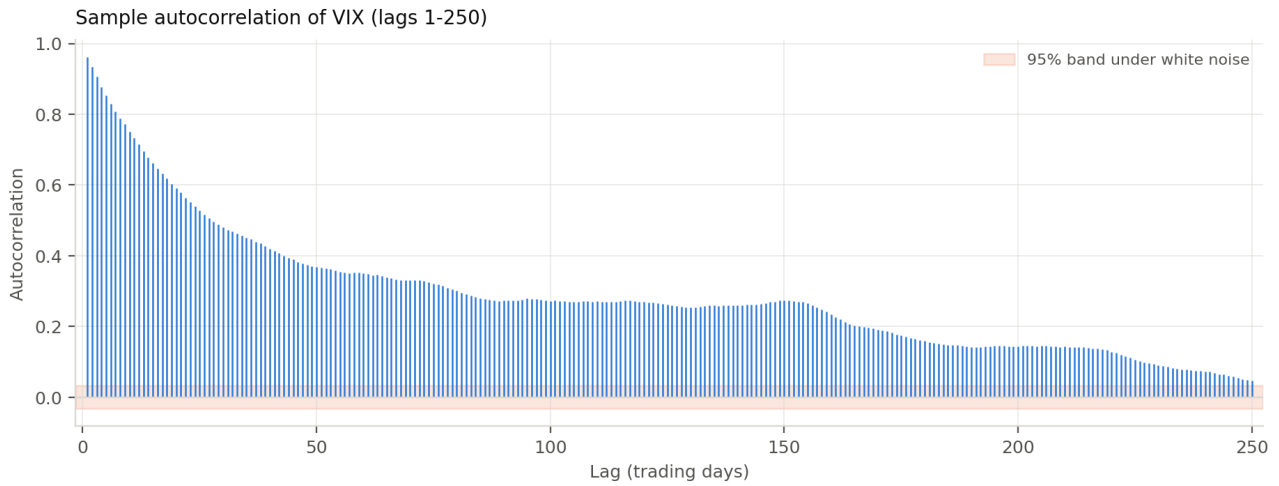


Figure 4: Sample autocorrelation of VIX to lag 250. The ACF remains outside the 95% white-noise band across the entire range.

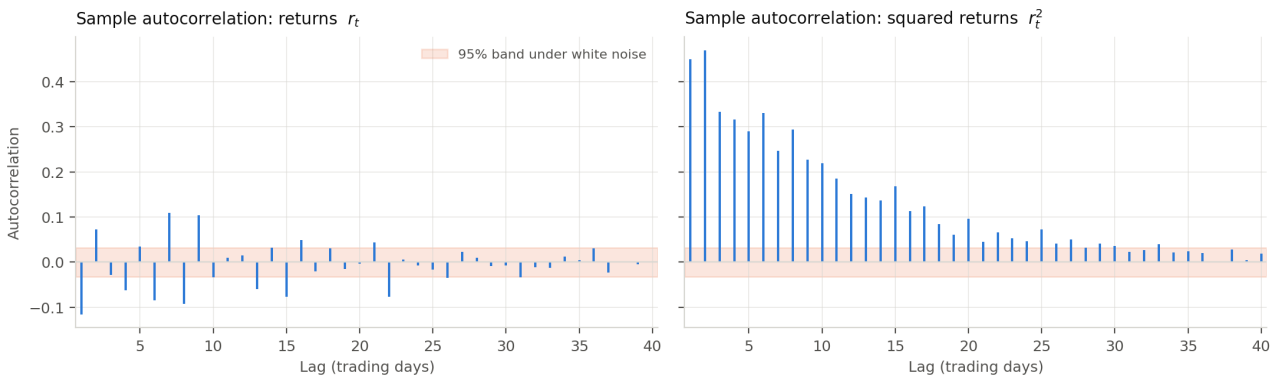


Figure 5: Sample autocorrelation of returns (left) and squared returns (right). Direction is unforecastable; magnitude is not.

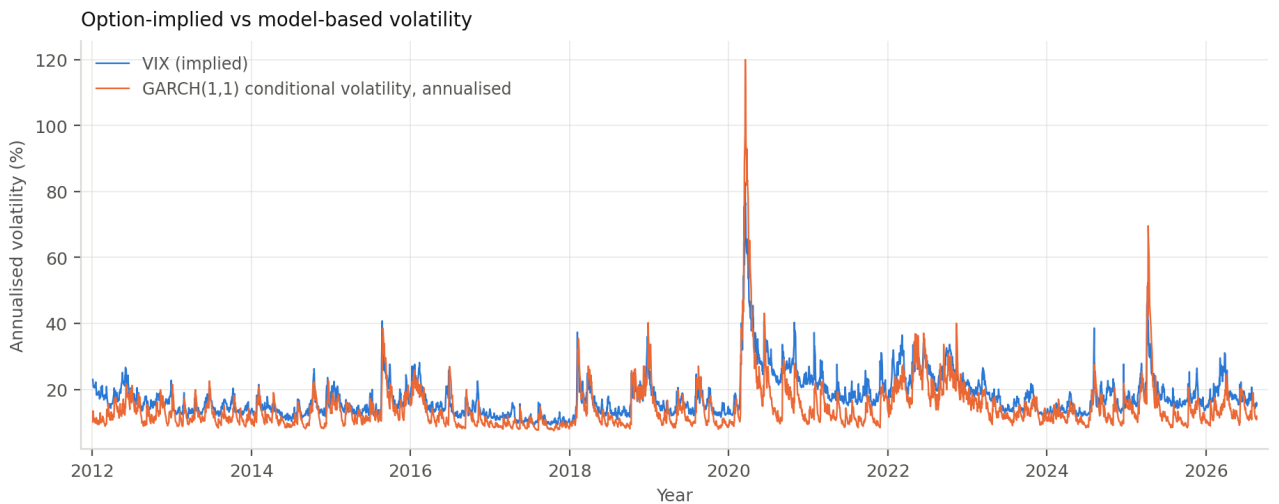


Figure 6: VIX against annualised GARCH(1,1) conditional volatility. Both series are in annualised volatility points and share a single axis.

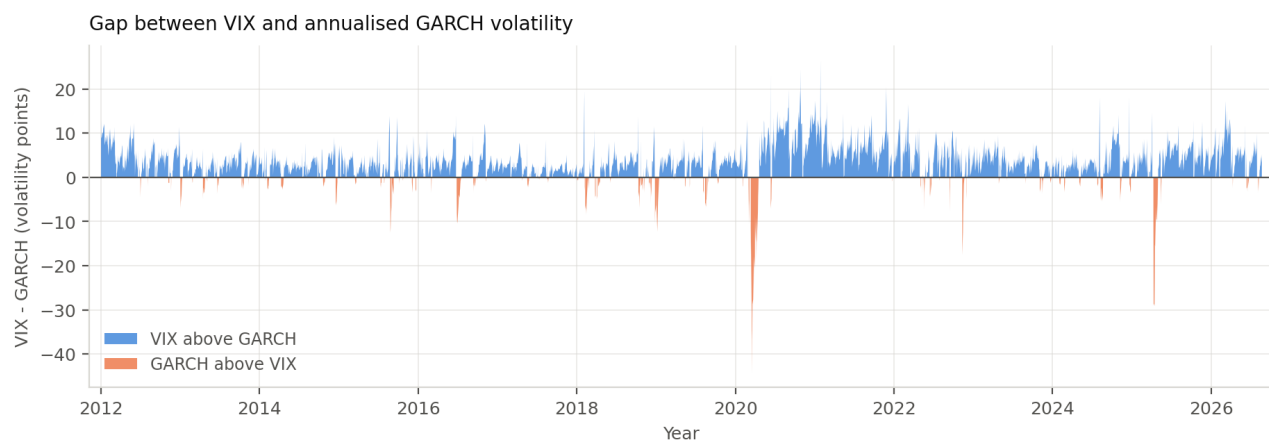


Figure 7: VIX minus annualised GARCH volatility. The predominance of positive values is the variance risk premium; the negative spikes follow extreme single-day returns.